

Trainings ▾

Preparatory Steps

Automotive with CM870

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Remove the water pump drive belt and tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Remove the refrigerant compressor drive belt. Refer to Procedure 009-052 in Section 9 (</qs3/pubsys2/xml/en/procedures/10/10-009-052-tr.html>).

Automotive with CM570

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).

- Remove the water pump and fan drive belt. Refer to Procedure 008-003 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-003.html).
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13 (/qs3/pubsys2/xml/en/procedures/10/10-013-005.html).
- Remove the alternator. Refer to Procedure 013-001 in Section 13 (/qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html).
- Remove the water pump belt tensioner bracket and belt tensioner. Refer to Procedure 008-080 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html).
- If necessary, remove the vibration dampers. Refer to Procedure 001-052 in Section 1 (/qs3/pubsys2/xml/en/procedures/10/10-001-052-tr.html).

Industrial CM570 and Power Generation without CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep children and pets away. If the coolant is not going to be reused, dispose of it in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).
- Remove the cooling fan and water pump belt. Refer to Procedure 008-003 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-003.html).
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13 (/qs3/pubsys2/xml/en/procedures/10/10-013-005.html).
- Remove the alternator. Refer to Procedure 013-001 in Section 13 (/qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html).
- If required, remove the damper assembly. Refer to Procedure 001-052 in Section 1 (/qs3/pubsys2/xml/en/procedures/10/10-001-052-tr.html).

Automotive With CM871

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

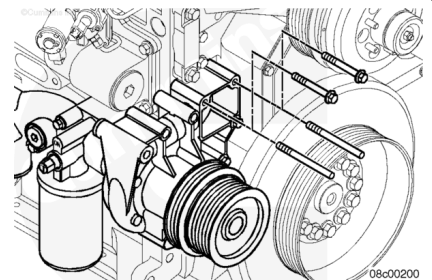
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Remove the water pump drive belt and tensioner. Refer to Procedure 008-080 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Remove the refrigerant compressor drive belt. Refer to Procedure 009-052 in Section 9 (</qs3/pubsys2/xml/en/procedures/10/10-009-052-tr.html>).
- Remove the water pump cartridge. Refer to Procedure 008-102 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-102-tr.html>).

Remove

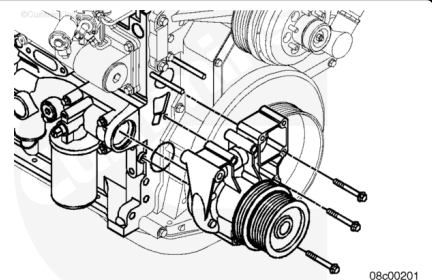
Automotive with CM870

Remove two water pump mounting capscrews. Install two guide studs.



Remove the remaining three capscrews, water pump, and o-ring seal.

Discard the o-ring seal.

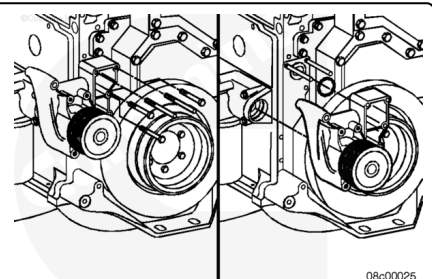


Automotive CM570 with One Piece Water Pump Design

Remove two water pump capscrews. Install guide studs.

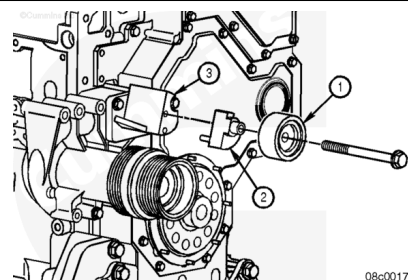
Remove remaining three capscrews, water pump, and water inlet o-ring seal.

Discard the o-ring seal.



Industrial CM570 with One Piece Water Pump Design and Power Generation without CM570

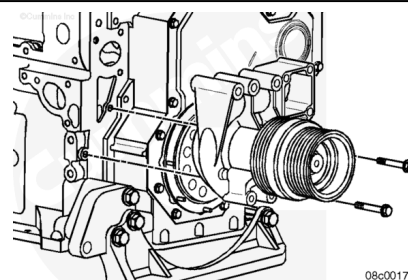
Remove the pulley (1), pilot adapter (2), and belt tensioner bracket (3).



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Remove the remaining water pump capscrews, and water inlet o-rings.

Discard the o-ring seals.



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Automotive With CM871

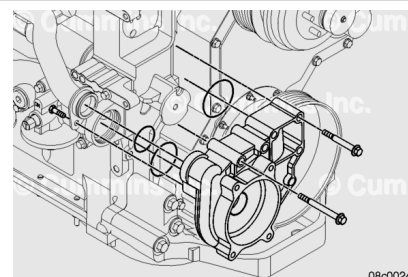
Remove the five water pump housing mounting capscrews.

Remove the two bracket mounting capscrews.

Remove the exhaust gas recirculation (EGR) tube outlet capscrew.

Remove the water pump housing and o-ring seals.

Discard the water inlet o-ring seal and EGR tube inlet and outlet o-ring seal.



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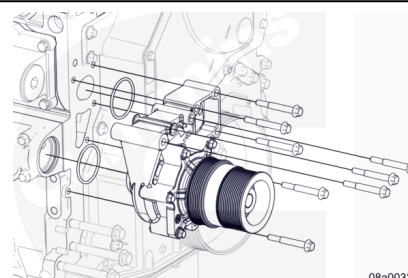
Automotive CM570 and Industrial CM570 with Two Piece Design Water Pump

Remove the five water pump housing mounting capscrews.

Remove the three bracket mounting capscrews.

Remove the water pump housing and o-ring seals.

Discard the water inlet o-ring seal

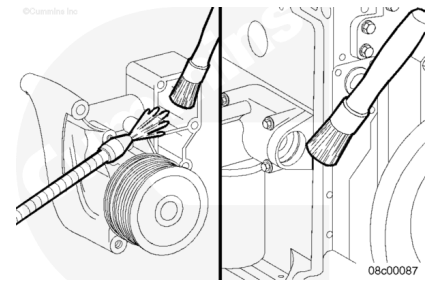


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Clean

One Piece Design, All Applications

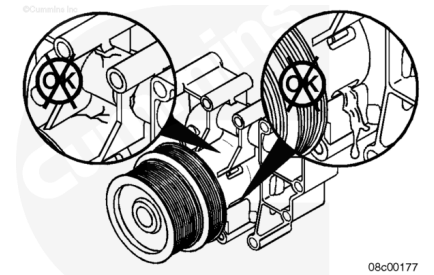
Clean the water pump and water inlet connections.



Inspect the water pump housing for cracks.

Inspect the water pump weep hole for indication of leaks.

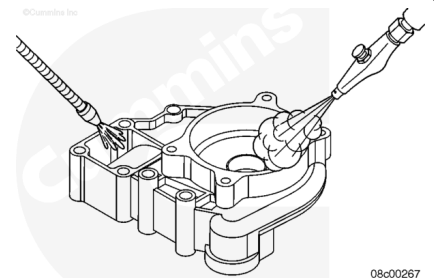
A streak or chemical buildup at the weep hole is **not** justification for water pump replacement. If a steady flow of coolant or oil is observed, recondition the existing water pump or replace the water pump with a new or rebuilt unit.



Automotive CM570/Industrial CM570 with Two Piece Design Water Pump and Automotive with CM871

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the water pump components. Use solvent. Dry with compressed air.

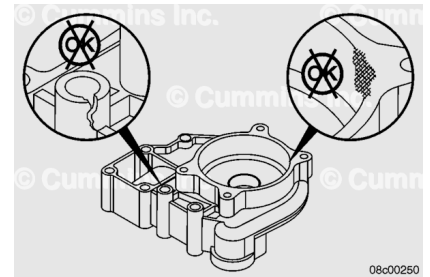
Inspect the water pump housing for cracks and damage.

Inspect the water pump inlet tube for out of roundness, damage, or leaks.

If the water pump housing is damaged, it must be replaced.

Continuous pitting deeper than 0.635 mm [0.025 in] or individual pits deeper than 2.54 mm [0.100 inch] are not acceptable.

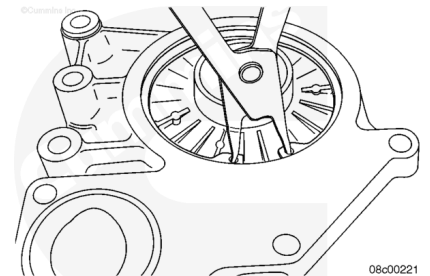
Replace the water pump housing if **not** within specifications.



Disassemble

⚠ WARNING ⚠

The retaining clip contains extreme amounts of potential energy. To reduce the possibility of personal injury from the clip slipping, place the water pump in a box prior to removing the retaining clip.



Note : This step **only** applies to the automotive CM570 and industrial CM570 with a one piece water pump design.

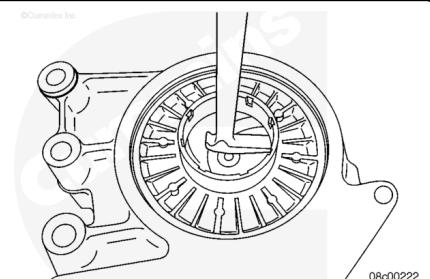
Place the water pump in a box.

Remove the retaining clip securing the impeller cover.

Remove and discard the water pump outlet o-ring.

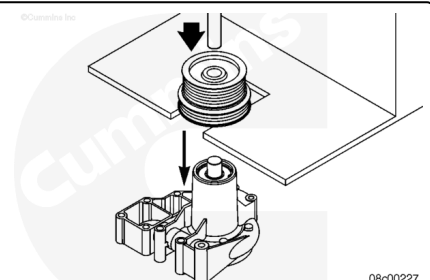
Use a heel bar to remove the impeller cover from the pump.

Discard the o-ring.



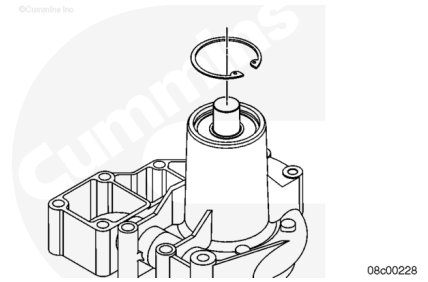
Support the water pump housing pulley in a press with the pulley facing up.

Press the pulley off of the water pump and shaft assembly.



Position the water pump with the pulley end facing up.

Remove the bearing bore retaining clip.

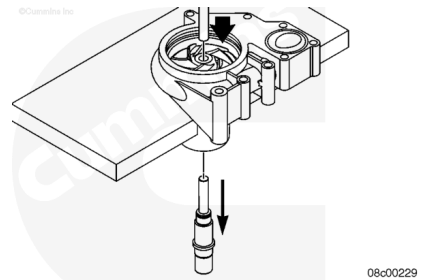


Support the water pump in a press with the impeller facing up.

Press the shaft out of the impeller and through the pump housing.

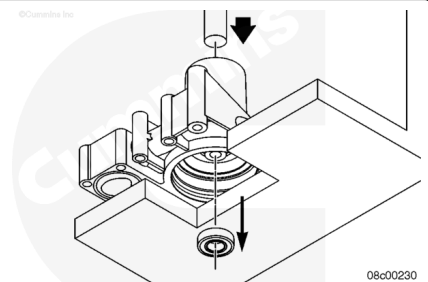
Remove the impeller from the housing.

Discard a phenolic impeller, retain a cast iron impeller.



Support the water pump in a press with the pulley end facing up.

Press the seal out of the housing.

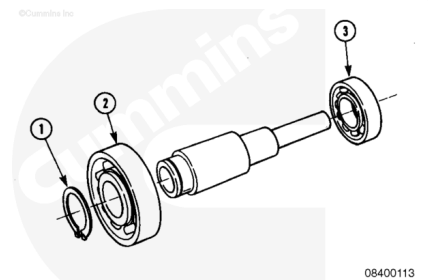


Note : Disassemble the shaft and bearings only if the seal is leaking or the bearing is damaged.

Remove the retaining ring (1).

Use water pump bearing separator, Part Number 3375326 to remove the:

- Front bearing (2)
- Rear bearing (3).



Clean and Inspect for Reuse

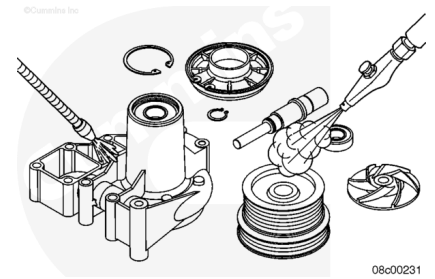
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

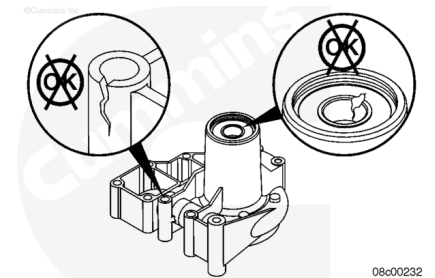
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : This step **only** applies to the automotive CM570 and industrial CM570 with a one piece water pump design.

Clean the water pump components. Use solvent. Dry with compressed air.

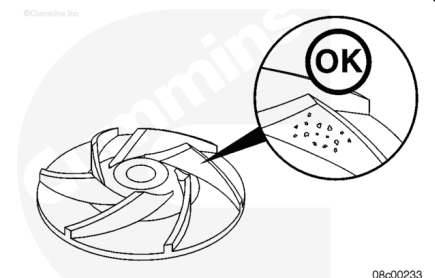
Inspect the water pump housing for cracks and damage.
If the water pump housing is damaged, it **must** be replaced.



Inspect the cast iron impeller for cracks, pitting, and other damage.

Continuous pitting deeper than 0.635 mm [0.025 in] or individual pits deeper than 2.54 mm [0.100 in] are **not** acceptable.

Replace the impeller if damaged or if pits are **not** within specifications.

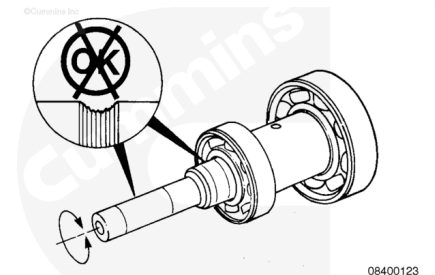


The water pump bearings are sealed bearings.

Spin the bearings to check for roughness.

Inspect the bearings for signs of grease purge.

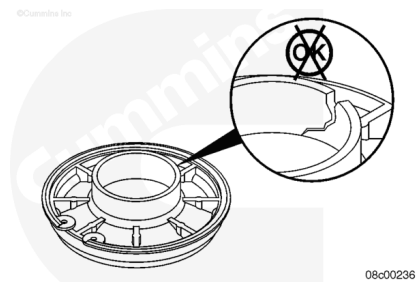
If the bearing seals show signs of grease purge or the bearing does **not** spin smoothly, it **must** be replaced.



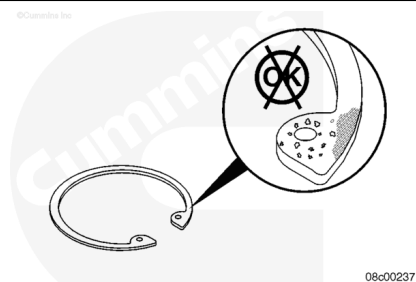
Inspect the impeller cover for cracks and other damage.

Scratching on the inside face is acceptable.

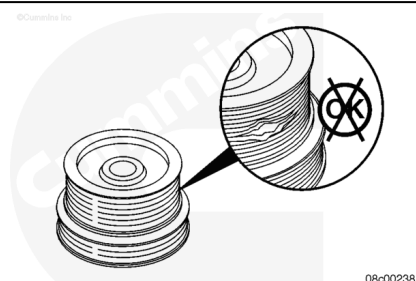
If there is any damage to the outer o-ring area, the impeller cover **must** be replaced.



Inspect the impeller cover retaining clip.
If the retaining clip is rusted or pitted, it **must** be replaced.



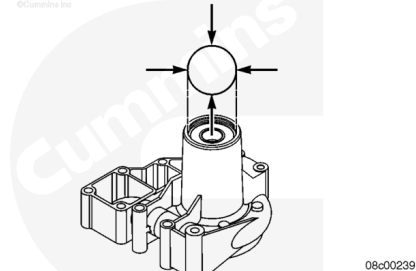
Inspect the pulley for cracks, wear, and damage.
Inspect the mating surface and pulley bore areas for damage.
If the pulley is damaged, it **must** be replaced.



Measure

Note : This step **only** applies to the automotive CM570 and industrial CM570 with a one piece water pump design.

Measure the inside diameter of the water pump housing large bearing bore.

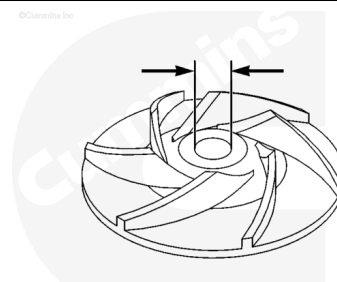


Water Pump Housing Large Bearing Bore Inside Diameter		
mm		in
71.988	MIN	2.834
72.018	MAX	2.835

If the water pump housing large bearing bore inside diameter is **not** within specifications, the water pump housing **must** be replaced.

Measure the inside diameter of the cast iron impeller.

Cast Iron Impeller Inside Diameter		
mm		in
15.850	MIN	0.624
15.875	MAX	0.625

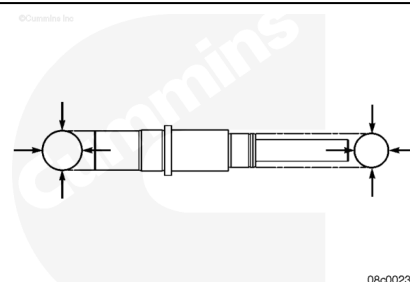


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If the cast iron impeller inside diameter is **not** within specifications, it **must** be replaced.

Measure the outside diameter of the water pump shaft.

Water Pump Shaft Diameter			
	mm		in
Small End	25.015	MIN	0.9848
	25.002	MAX	0.9843
Large End	29.015	MIN	1.1423
	29.002	MAX	1.1418



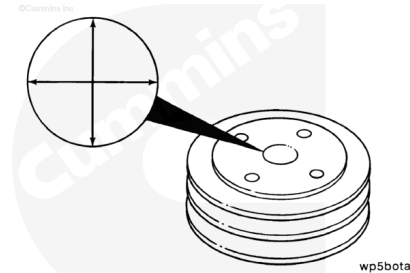
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Measure the outside diameter of the larger end of the shaft.

If the shaft is **not** within specifications, it **must** be replaced.

Measure the inside diameter of the pulley bore.

Pulley Bore Inside Diameter		
mm		in
28.936	MIN	1.139
29.964	MAX	1.140



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If the pulley bore is **not** within specifications, the pulley **must** be replaced.

Assemble

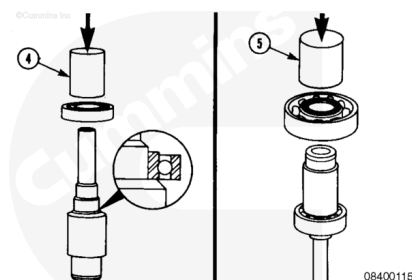
⚠ CAUTION ⚠

The mandrel must make contact on the inner race of the bearing to prevent damage to the bearing.

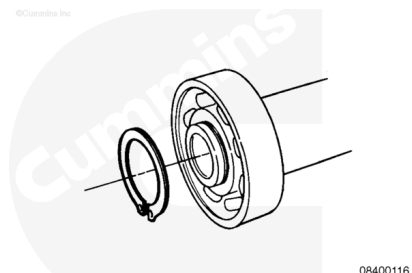
Note : This step only applies to the automotive CM570 and industrial CM570 with a one piece water pump design.

Support the shaft in an arbor press.

Use an arbor press with water pump mandrel, Part Number ST-658 (4), or equivalent, to install the bearings.



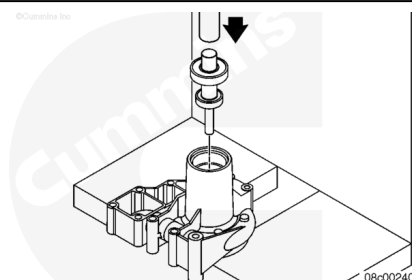
Install the retaining clip with the beveled side of the clip facing away from the bearing.



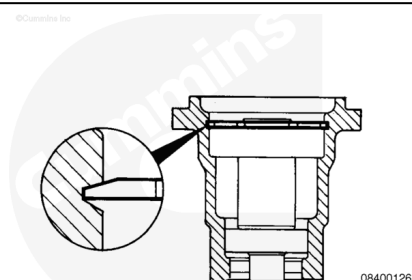
Support the water pump housing on a table with the pulley end facing up.

Install the shaft and bearing assembly into the bearing bore with the smaller bearing end of the assembly entering first.

If the shaft does **not** drop into position from its weight, the assembly can be pressed into place.



Install the retaining clip with the beveled side of the clip facing away from the bearing.



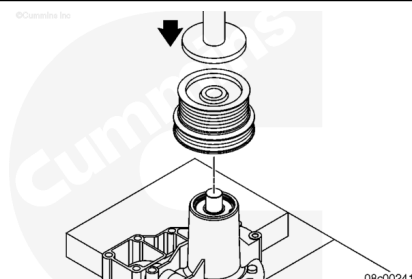
⚠ CAUTION ⚠

To reduce the possibility of damage to the water pump housing, make sure the impeller end of the shaft is supported before pressing on the pulley.

Support the impeller end of the shaft in a press.

Press on the pulley until it is flush with the end of the shaft.

Make sure the pulley rotates freely.

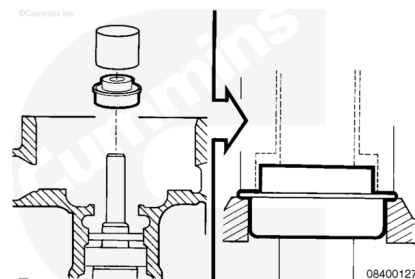


Press the water pump seal onto the shaft.

Measure the water pump seal installed height.

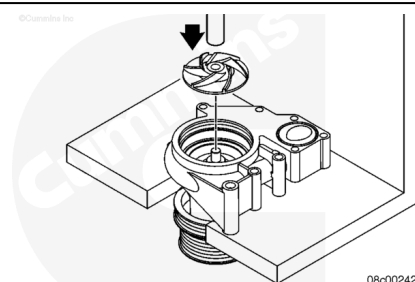
Water Pump Seal Installed Height		
mm		in
10.919	MIN	0.430
11.679	MAX	0.460

If the water pump seal installed height is **not** within specifications, the seal **must** be removed and installed again.



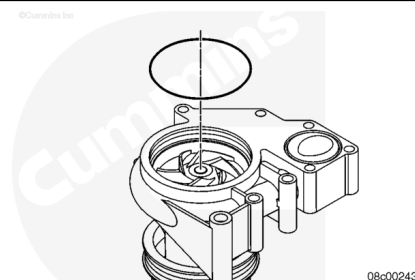
Support the water pump in a press with the pulley down.

Press the impeller, vane side up, onto the shaft until it is flush with the end of the shaft.



⚠ CAUTION ⚠

Do not use petroleum based lubricants. Petroleum based lubricants cause the o-ring seals to crack, leading to a coolant leak.



⚠ CAUTION ⚠

Make sure the o-ring is seated properly. If the o-ring is not seated properly it can be damaged during impeller cover installation.

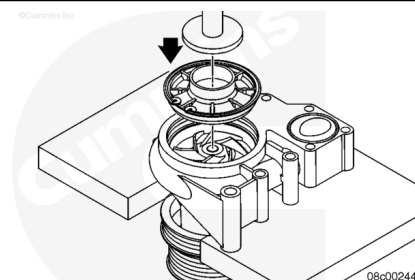
Lubricate the impeller cover o-ring. Use soapy water.

Install the cover o-ring into the water pump housing.

⚠ CAUTION ⚠

Do not use petroleum lubricant on the impeller cover, o-ring damage can result from the use of petroleum lubricant.

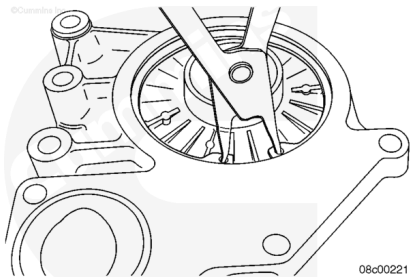
Lubricate the outside diameter of the impeller cover and the o-ring bore. Use soapy water.



Support the water pump in a press.
Press the impeller cover into the bore until seated.

⚠ WARNING ⚠

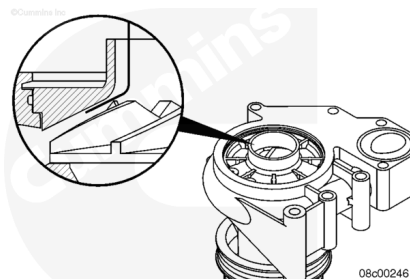
The retaining clip contains extreme amounts of potential energy. To reduce the possibility of personal injury from the clip slipping, place the water pump in a box prior to installing the retaining clip.



Place the water pump in a box.
Install the retaining clip.

Measure the impeller clearance between the impeller and impeller cover. Use a feeler gauge.

Impeller Clearance		
mm		in
0.29	MIN	0.011
1.24	MAX	0.049



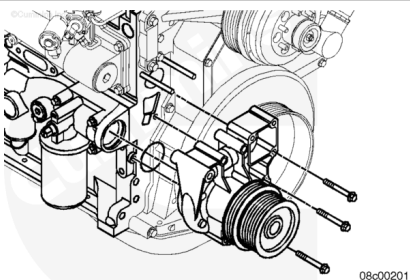
Install

Automotive with CM870

⚠ CAUTION ⚠

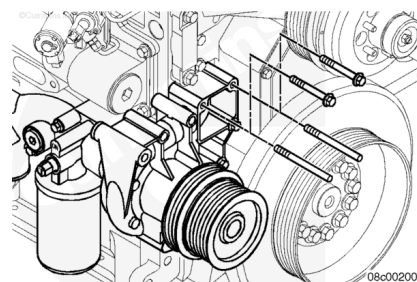
Do not use petroleum based lubricants. Petroleum based lubricants cause the o-ring seals to crack, leading to a coolant leak.

Lubricate and install the new o-ring seal.
Install the two guide studs.
Install the water pump and three capscrews.



Remove the two guide studs. Install the remaining mounting capscrews.
Tighten the capscrews.

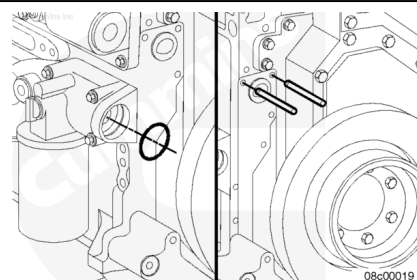
Torque Value: 72 n•m [53 ft-lb]



Automotive with CM570

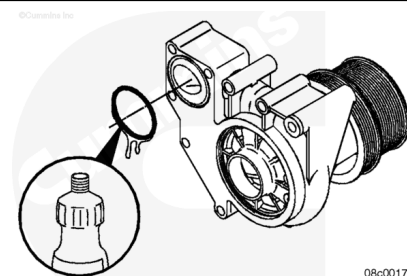
Install a new o-ring seal in the water inlet.

Install the two guide studs.



⚠ CAUTION ⚠

Do not use petroleum based lubricants. Petroleum based lubricants cause the o-ring seals to crack, leading to a coolant leak.



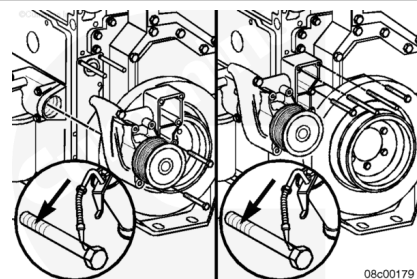
Install a new o-ring seal in the water pump outlet groove.

Apply a thin layer of non-petroleum-based lubricant, such as vegetable oil, to both o-ring seals.

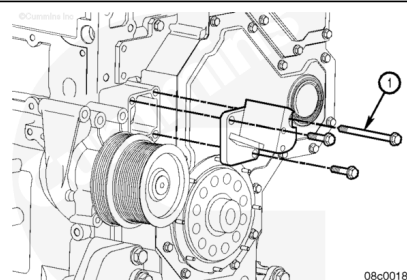
Use clean engine oil to lubricate the threads and bottom of the capscrew heads of all the water pump capscrews.

Install the water pump and two lower capscrews.

Tighten the capscrews to hold the pump in place.



Note : Some configurations do not use the middle aluminum belt tensioner bracket shown. Refer to Procedure 008-088 in Section 8 (</qs3/pubsys2/xml/en/procedures/10/10-008-088.html>) for the appropriate tensioner bracket installation.



Install the belt tensioner bracket, if equipped, and capscrews in the location shown.

Finger tighten all capscrews.

Tighten the water pump mounting capscrews in the sequence and values shown.

Torque Value:

Capscrews 1, 2, and 3

1. 58 n•m [43 ft-lb]

Torque Value:

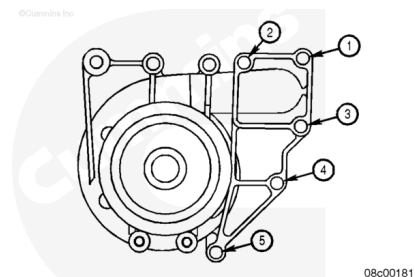
Capscrews 4 and 5

1. 72 n•m [53 ft-lb]

Torque Value:

All Capscrews, Engines Without Belt Tensioner Bracket

1. 72 n•m [53 ft-lb]

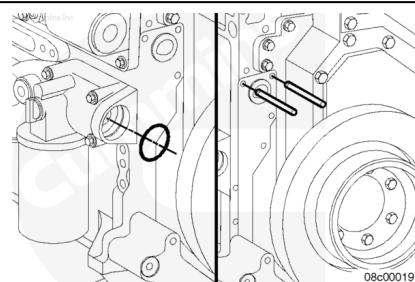


08c00181

Industrial CM570 and Power Generation without CM570

Install a new o-ring seal in the water inlet.

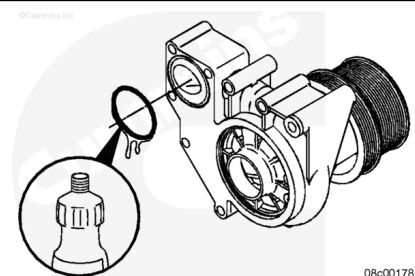
Install the two guide studs.



08c00019

⚠ CAUTION ⚠

Do not use petroleum based lubricants. Petroleum based lubricants cause the o-ring seals to crack, leading to a coolant leak.



08c00178

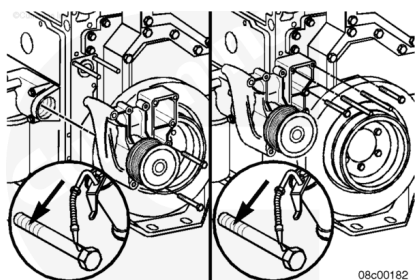
Install a new o-ring seal in the water pump outlet groove.

Apply a thin layer of non-petroleum-based lubricant, such as vegetable oil, to both o-ring seals.

Use clean engine oil to lubricate the threads and bottom of the capscrew heads of all the water pump capscrews.

Position the water pump on the engine.

Install the bottom two capscrews finger-tight.



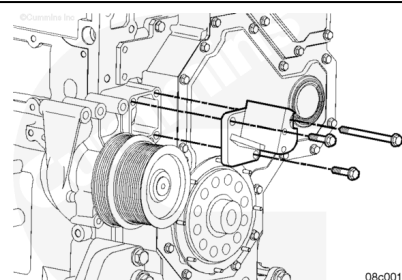
08c00182

Note : Some configurations do not use the middle aluminum belt tensioner bracket. Refer to Procedure 008-088 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-088-tr.html) for the appropriate tensioner bracket installation.

Note : Early engines that did not have a slotted hole on the middle aluminum belt tensioner idler bracket required the vibration dampers to be removed to remove and install the water pump.

Install the belt tensioner bracket, if equipped, and the capscrews in the locations shown.

Finger tighten all capscrews.



08c00183

Tighten the water pump mounting capscrews in the sequence and values shown.

Torque Value:

Capscrews 1, 2, and 3

1. 58 n•m [43 ft-lb]

Torque Value:

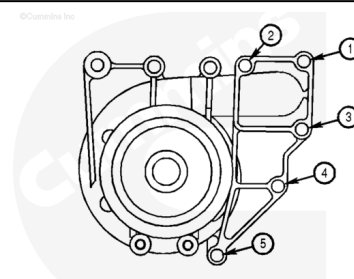
Capscrews 4 and 5

1. 72 n•m [53 ft-lb]

Torque Value:

All Capscrews, Engines Without Belt Tensioner Bracket

1. 72 n•m [53 ft-lb]



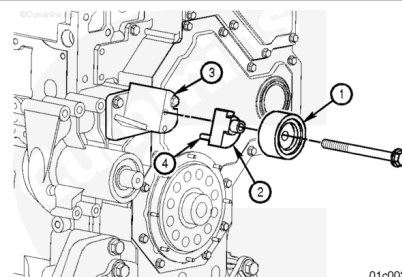
08c00181

Align the idler pulley (1) and the shaft of the pilot adapter (2) with the belt tensioner bracket (3) using the guide pin on the pilot adapter (4), if equipped.

Install the belt idler pulley and shaft on the belt idler bracket using one washer and capscrew.

Tighten the capscrew.

Torque Value: 30 n•m [25 ft-lb]



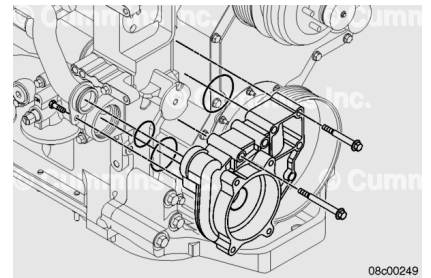
01c00282

Install the dust seal.

Automotive With CM871

⚠ CAUTION ⚠

Do not use petroleum based lubricants, petroleum based lubricants cause the o-ring to crack or swell, leading to a coolant leak.



Install a new water pump inlet o-ring seal and a new EGR tube outlet o-ring seal.

Install the five water pump housing capscrews.

Tighten the capscrews.

Torque Value: 72 n•m [53 ft-lb]

Install the two bracket capscrews.

Tighten the capscrews.

Torque Value: 30 n•m [25 ft-lb]

Install the EGR tube outlet capscrew.

Tighten the capscrew.

Torque Value: 30 n•m [25 ft-lb]

Finishing Steps

Automotive with CM870

- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html>).
- Install the water pump drive belt and tensioner. Refer to Procedure 008-080 (</qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html>).
- Install the refrigerant compressor drive belt. Refer to Procedure 009-052 (</qs3/pubsys2/xml/en/procedures/10/10-009-052-tr.html>).

Automotive with CM570

- Install the vibration dampers, if removed. Refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/10/10-001-052-tr.html>).
- Install the alternator. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html>).

- Install the water pump belt tensioner bracket and belt tensioner. Refer to Procedure 008-080 (/qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html).
- Install the alternator drive belt. Refer to Procedure 013-005 (/qs3/pubsys2/xml/en/procedures/10/10-013-005.html).
- Install the water pump drive belt. Refer to Procedure 008-003 (/qs3/pubsys2/xml/en/procedures/10/10-008-003.html).
- Fill the engine with heavy-duty coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).
- Start the engine and check for leaks.

Industrial CM570 and Power Generation without CM570

- Install the damper assembly, if removed. Refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/10/10-001-052-tr.html).
- Install the alternator. Refer to Procedure 013-001 (/qs3/pubsys2/xml/en/procedures/10/10-013-001-tr.html).
- Install the alternator drive belt. Refer to Procedure 013-005 (/qs3/pubsys2/xml/en/procedures/10/10-013-005.html).
- Install the fan and water pump drive belt. Refer to Procedure 008-003 (/qs3/pubsys2/xml/en/procedures/10/10-008-003.html).
- Fill the engine with heavy-duty coolant. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).
- Start the engine and check for leaks.

Automotive With CM871

- Install the water pump cartridge. Refer to Procedure 008-102 (/qs3/pubsys2/xml/en/procedures/10/10-008-102-tr.html).
- Install the refrigerant compressor drive belt. Refer to Procedure 009-052 (/qs3/pubsys2/xml/en/procedures/10/10-009-052-tr.html).
- Install the water pump drive belt and tensioner. Refer to Procedure 008-080 (/qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html).
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).
- Operate the engine and check for leaks.